

AIS Graph Analytics

From vessel positions to port-network insights.

Streaming ingestion • Historical processing • Graph analytics

48.7M

historical position rows*

183

unit tests passed

4 / 4

GitHub CI jobs passed

* Canonical rows before 24 Sep 2026, measured during acceptance.

Raw telemetry needs analytical context.

01 Observe

Collect live vessel positions and retain historical observations for repeatable analysis.

02 Interpret

Infer bounded port stays and consecutive observed movements between ports.

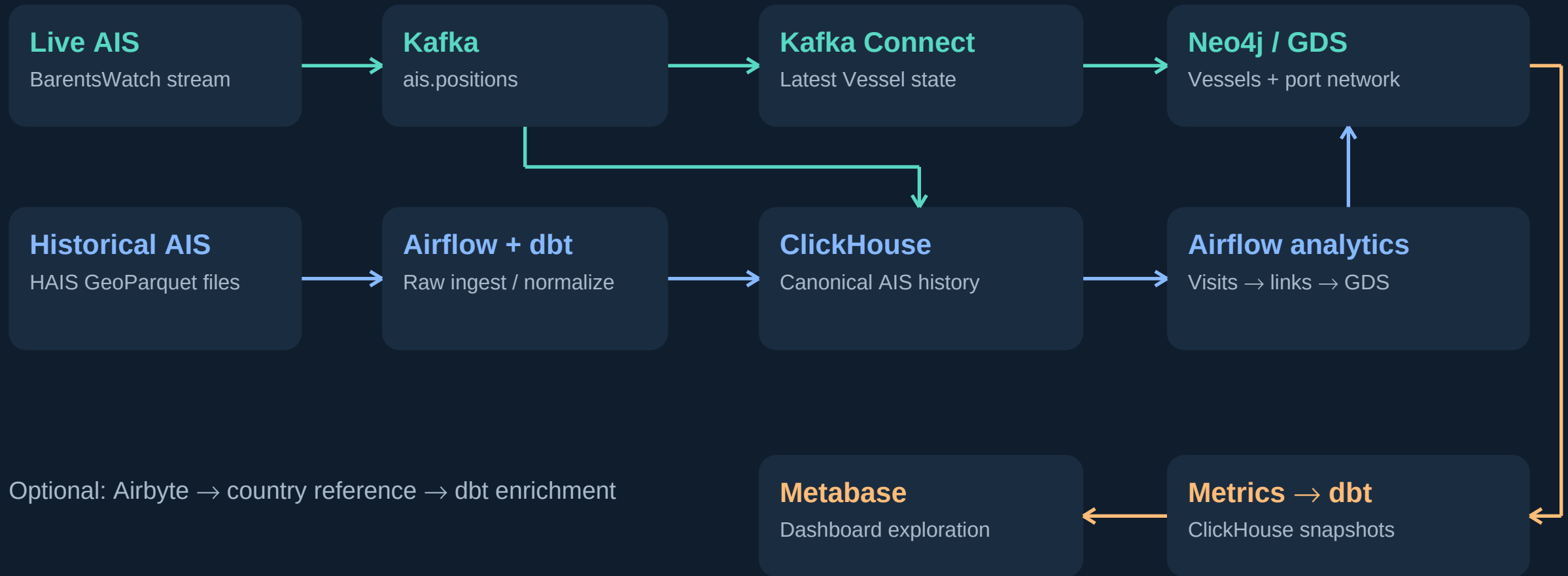
03 Explain

Rank connected ports, identify communities, and explore metrics in Metabase.

Core question

How does observed vessel activity form a network of ports?

Two ingestion paths. One analytical workflow.



Docker Compose local runtime • Airflow orchestrates batch analytics

Different data paths serve different needs.

Live stream

Continuous producer with token refresh and reconnect handling.

Kafka fans out to the current vessel graph and analytical position history.

Historical files

Airflow loads HAIS files with audit records.

dbt normalizes observations into the canonical ClickHouse table.

Identity matters

MMSI identifies vessels; UTC windows identify analytical runs.

A daily chain turns positions into graph metrics.

1 Detect visits

Positions → qualified stays



2 Publish links

Consecutive visits → edges



3 Run GDS

PageRank + Louvain



4 Export

Versioned metrics → BI

Repeatable windows

Explicit UTC boundaries and shared run identity connect inputs, visits, graphs, and exports.

Controlled progression

The next day waits for successful export. Snapshot checks block stale or inconsistent outputs.

Relationships add context to port activity.



Schema illustration — not a specific vessel journey

PageRank

Relative importance in the observed movement network.

Louvain communities

Groups of ports with stronger internal connectivity.

One day of data, traced through the stack.

Processing window: 23 Sep 2026, 00:00–24:00 UTC

2.94M

Source positions

2,941,803 inputs processed

2,696

Detected visits

Daily analytical visit records

51

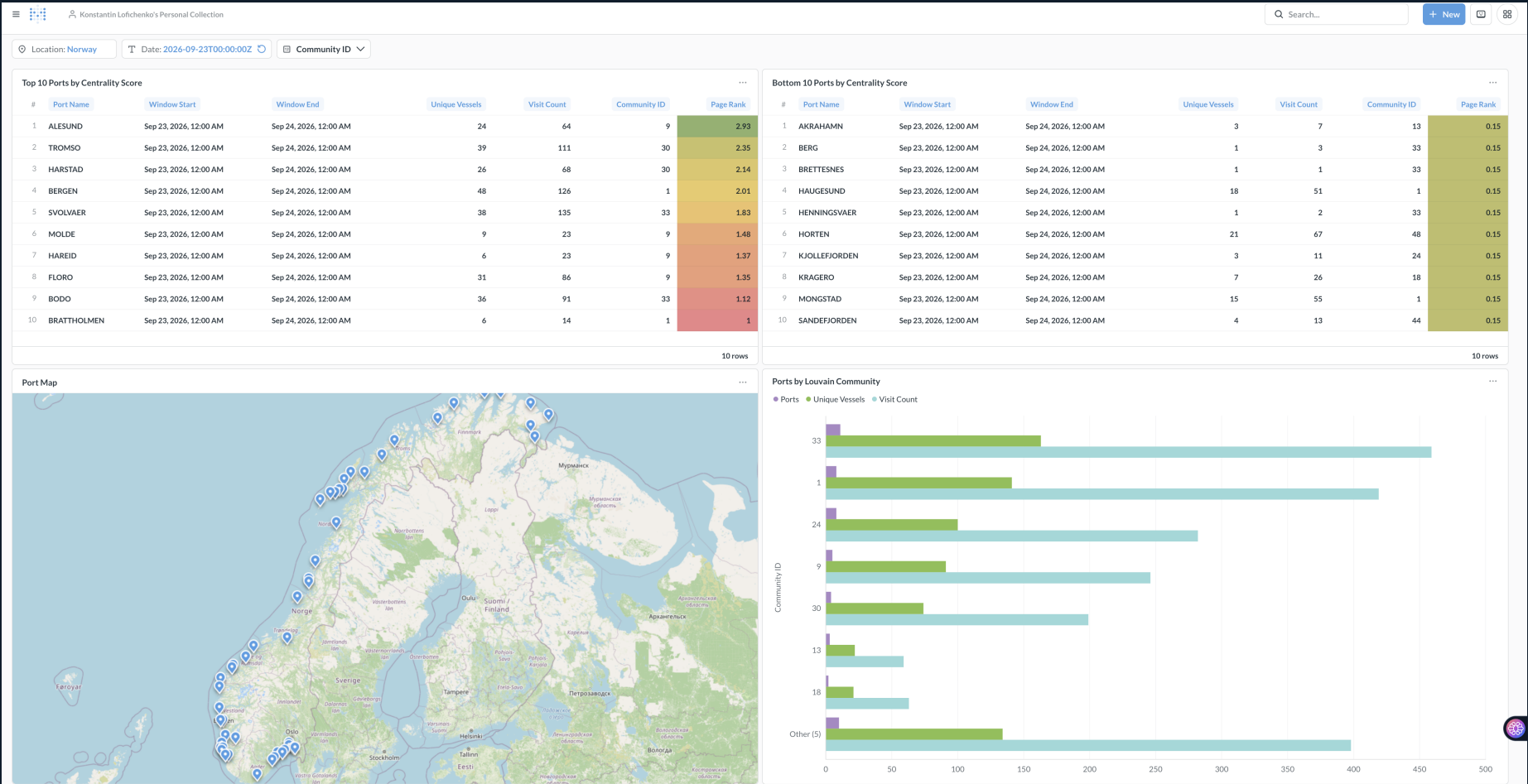
Port metric rows

Country-enriched graph output

Metabase dashboard visually confirmed by the project owner.

Observed counts document this run; they are not a performance benchmark.

The analytical output is ready to explore.



PageRank tables • Port map • Louvain community summary

Correctness before convenience.

Validate snapshot identity

Run/window metadata and graph checks prevent stale metrics from being exported.

Handle empty outcomes

A valid zero-edge snapshot is different from an incomplete or missing publication.

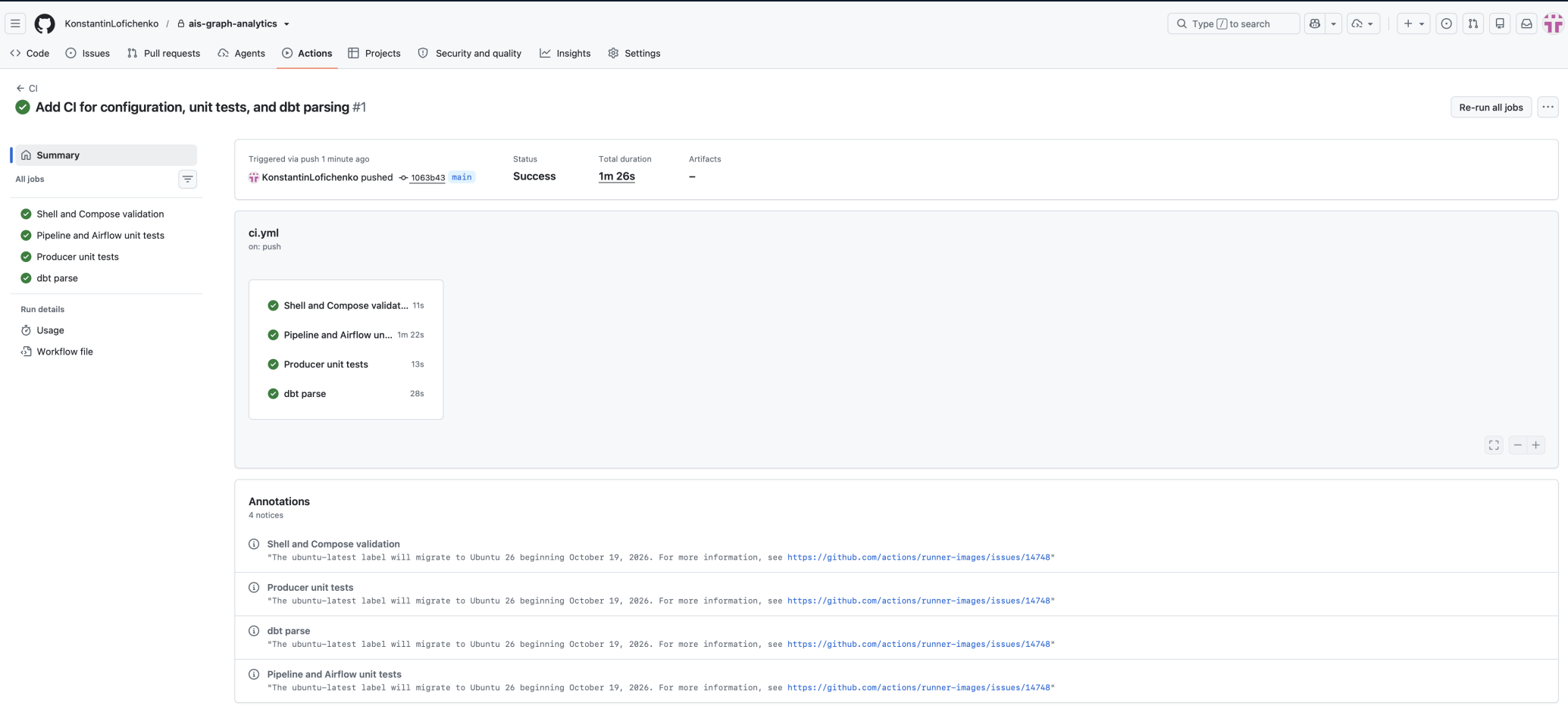
Verify restart persistence

Counts, analytical checksums, constraints, and 136 Airflow run records survived stop/start.

Automate repeatable checks

183 unit tests passed locally; CI also validates shell, Compose, and dbt parsing.

The first GitHub CI run passed.



A working portfolio platform—with clear limits.

Completed

Live and historical data paths

Port network + GDS + Metabase

Local runtime acceptance

Restart persistence + GitHub CI

Next engineering work

Stays spanning midnight

Scale beyond bounded in-memory batches

Backup / restore validation — deferred

Fresh-install validation — deferred

From ingestion to a tested analytical product.

Data engineering

Streaming + batch ingestion, canonical modeling, and orchestration.

Graph analytics

Observed port movements, PageRank, communities, and BI delivery.

Engineering discipline

Lineage checks, reproducible runs, automated tests, and explicit limits.

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